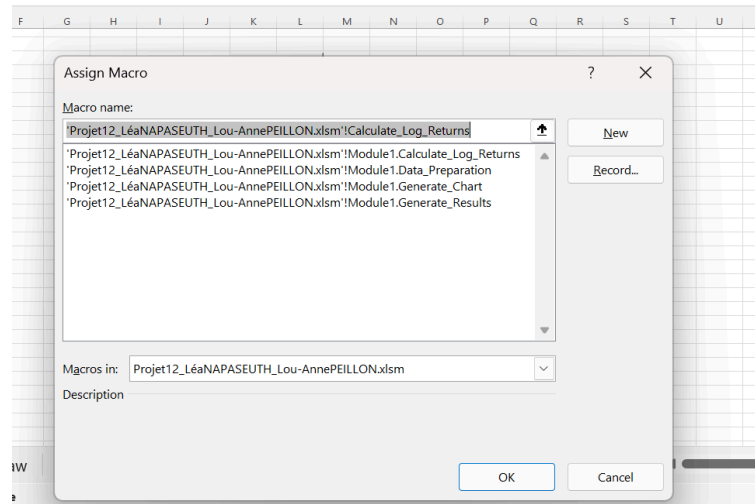


EXCEL VBA PROJECT 12 : CHF and Gold

0. Preliminary

To enable the spreadsheet's functionality, please assign the 4 macros to each buttons.



1. Introduction and objectives

The purpose of this project is to analyse the statistical relationship between two major financial assets considered as safe-haven assets : Gold and CHF. In a context of unstable economy, geopolitics issues, these assets are the key for investors to protect themselves from loss. Our analysis aims to verify their measurable correlation between the variation of price of the gold and CHF. Our study relies on historical data, descriptive statistics, correlation and linear regression under Excel and VBA.

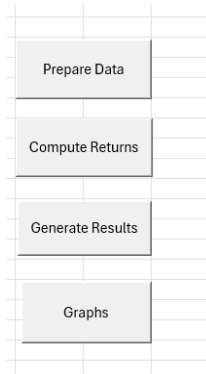
2. Data and sources

We use Yahoo Finance source with Gold/USD as $GC=F$ (Gold Futures) and USD/CHF as $USDCHF=X$. Our studies period covers a few years (2020 to 2025) and is enough to get a pertinent analysis.

A	B	C	D	E
Date Gold	Price Gold		Date USDCHE	Price USDCHE
22/12/2020	1866.5999755859375		22/12/2020	0.8854799866676331
23/12/2020	1874.699951171875		23/12/2020	0.8878600001335144
24/12/2020	1879.9000244140625		24/12/2020	0.8878999948501587
25/12/2020	1877.199951171875		25/12/2020	0.8909000158309937
29/12/2020	1879.699951171875		28/12/2020	0.8894299864768982
30/12/2020	1891.0		29/12/2020	0.8883000016212463
31/12/2020	1893.0999755859375		30/12/2020	0.8841000199317932
04/01/2021	1944.699951171875		31/12/2020	0.8815799951553345
05/01/2021	1952.699951171875		01/01/2021	0.8981000185012817
06/01/2021	1906.9000244140625		04/01/2021	0.8830999732017517
07/01/2021	1912.300048828125		05/01/2021	0.8812400102615356
08/01/2021	1834.0999755859375		06/01/2021	0.8781800270080566
11/01/2021	1849.5999755859375		07/01/2021	0.8780699968338013
12/01/2021	1842.9000244140625		08/01/2021	0.8851500153541565
13/01/2021	1853.5999755859375		11/01/2021	0.8876399993896484
14/01/2021	1850.300048828125		12/01/2021	0.8896200060844421
15/01/2021	1829.300048828125		13/01/2021	0.8859000205993652
19/01/2021	1839.5		14/01/2021	0.886900007724762
20/01/2021	1865.9000244140625		15/01/2021	0.8882499933242798

3. VBA Methods

The *Home* sheet is designed as the main user interface of the Excel/VBA tool. It allows the user to run each step of the analysis without directly interacting with the calculation sheets or the VBA code.



- The **Prepare Data** button initiates the data preparation phase. It cleans the raw data imported from Yahoo Finance, harmonizes date and numerical formats, synchronizes the trading calendars of Gold and CHF and converts the USD/CHF exchange rate into CHF/USD in order to express both assets in the same currency. Gold is quoted in USD, while the Swiss franc is generally quoted as USD/CHF. To make the two series comparable, a conversion was made : $\text{CHF/USD} = 1/ \text{USD/CHF}$. This transformation allows both assets to be expressed in the same currency (USD) and avoids an artificially negative correlation that would fake our results.

	C
	CHF_Price_in_USDR
3	1,129331001
4	1,126303696
5	1,126252963
6	1,122460413
7	1,125745805
8	1,131093742
9	1,13432701
10	1,132374624
11	1,134764637
12	1,138718679
13	1,138861371
14	1,129752
15	1,12658285
16	1,12407544
17	1,128795549
18	1,127522823
19	1,125809184
20	1,122460413
21	1,126126112
Raw	Calculs

- The **Compute Returns** button calculates the logarithmic returns from the cleaned price series. In finance, in general, the analysis is not based on raw prices but on logarithmic returns, defined as:

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right)$$

```
'2nd button : compute Returns
Sub Calculate_Log_Returns()
    Dim ws As Worksheet
    Dim L As Long, lastRow As Long

    Set ws = Worksheets("Calculs")
    lastRow = ws.Cells(ws.Rows.Count, 1).End(xlUp).Row

    For L = 3 To lastRow
        'LN(Price T / Price T-1)
        If ws.Cells(L, 2).Value > 0 And ws.Cells(L - 1, 2).Value > 0 Then
            ws.Cells(L, 4).Value = Application.WorksheetFunction.Ln(ws.Cells(L, 2).Value / ws.Cells(L - 1, 2).Value)
        End If

        If ws.Cells(L, 3).Value > 0 And ws.Cells(L - 1, 3).Value > 0 Then
            ws.Cells(L, 5).Value = Application.WorksheetFunction.Ln(ws.Cells(L, 3).Value / ws.Cells(L - 1, 3).Value)
        End If
    Next L

    ws.Range("D3:E" & lastRow).NumberFormat = "0.00%"
    MsgBox "Step 2 completed: Log returns calculated."
End Sub
```

It gives us a better stationarity of the series, time additivity and is more relevant in comparison of relative variations.

- The **Generate Results** button performs the core quantitative analysis. It automatically computes: descriptive statistics (mean, volatility, minimum and maximum), the correlation between Gold and CHF returns and the parameters of a

linear regression model, including the beta coefficient and the coefficient of

A	B
Statistic indicators	Values
Mean Gold	0,0007
Volatility Gold	0,0100
Corelation Or vs CHF	0,0691
Beta (Gold/CHF)	0,1368
R ²	0,004769018

determination (R²).

- **The Graphs** button generates the visual outputs of the analysis. The regression implemented in the project is a simple linear regression defined as:

$$r_t^{Gold} = \alpha + \beta \cdot r_t^{CHF} + \varepsilon_t$$

where R_t^{Gold} is the return of Gold at time t , R_t^{CHF} is the logarithmic return of CHF/USD at time t , α is the intercept, β is the regression coefficient (beta) and ε_t is the error term.

4. Excel Methods

We do exactly the same step but in Excel. Logarithmic returns, calculated as:

D3	:	$\times$	$\checkmark$	f_x	=LN(B3/B2)
	A	B	C	D	E
1	Date	GoldPrice	CHF/USD Price	Gold Return	CHF return
2	22/12/2020	1866,6	1,129331001		
3	23/12/2020	1874,7	1,126303696	0,00433004	-0,00268422
4	24/12/2020	1879,9	1,126252963	0,00276998	-4,5045E-05
5	25/12/2020	1877,2	1,122460413	-0,00143732	-0,00337309
6	29/12/2020	1879,7	1,125745805	0,00133088	0,00292268
7	30/12/2020	1891	1,131093742	0,00599363	0,004739325

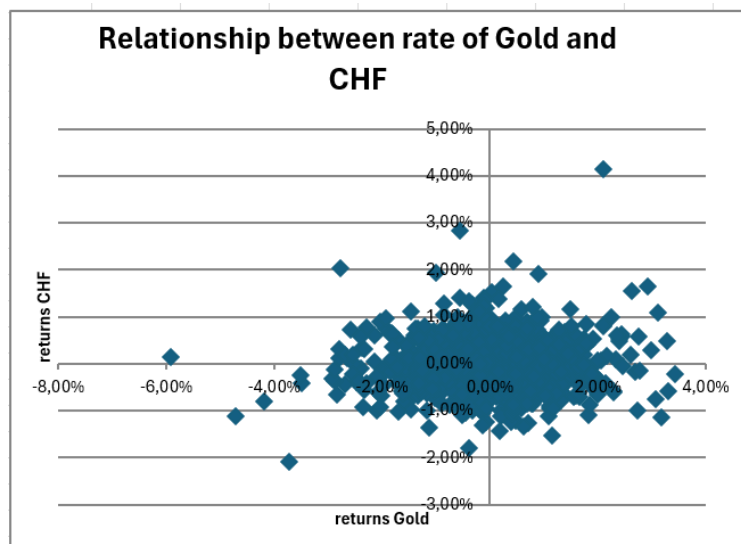
We compute descriptive statistics : mean with AVERAGE, volatility with STDEV , minimum with MIN and maximum with MAX. Then, for correlation between Gold and CHF returns we use the Excel CORREL function. Finally, for linear regression we use SLOPE, INTERCEPT and RSQ exactly as presented during lectures. We obtain these results in the following table :

Metrics	Gold Excel	CHF Excel
Mean	0,000690741	8,5E-05
Volatility	0,010019933	0,005056
Min	-0,05906204	-0,02082
Max	0,034367644	0,041397
Correlation	0,069058075	
Beta	0,13684775	
Intercept	0,000679111	
R ²	0,004769018	

5. Results and Comparison

Metrics	Comparison	
	EXCEL	VBA
correlation	0,06905808	0,06906
beta	0,13684775	0,13685
R ²	0,00476902	0,00477

This is the table of results and they are the same in VBA and Excel.



The results show a really weak positive link between gold and CHF returns. Gold is way more volatile than the Swiss Franc, even though both assets have average returns that are pretty much zero. The correlation is pretty low (around 0.07), which basically means their short-term moves don't have much to do with each other. The regression backs this up, to the beta is positive but tiny, and the R² is so low that CHF returns barely explain anything when it comes to gold's performance.

6. Conclusion

This project involved developing an automated Excel and VBA tool to analyze the relationship between Gold and the Swiss Franc. The results highlight a positive but however limited correlation. Also, there was a low explanatory power within the framework of a simple

linear model. Gold and the Swiss franc maintain their status as safe-haven assets, yet they respond to different market dynamics.

Our study was limited by the small period of data, a simplified linear model and an absence of explanatory macroeconomic variables. However, to improve our project, we could roll correlation analysis, extend to other safe-haven currencies (JPY) or even compute Non-linear or multivariate models.